

Fig. 1: MOS capacitor

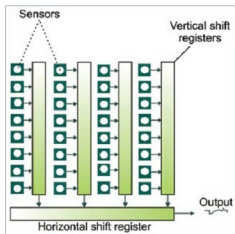


Fig. 2: CCD array

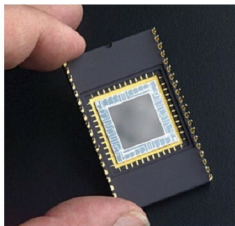


Fig. 3: Packaged CCD chip

computer. This digital bit pattern then represents the image.

The charge held in bins corresponding to different pixels is read out, converted into equivalent analogue voltages and then digitised with the help of an analogue-to-digital converter. Charge on a CCD is shifted in two directions: parallel (or horizontal) and serial (or vertical). While parallel shift occurs from right to left, serial shift is performed from top to bottom and directs charge packets to the measurement electronics. One way to make the read-out process faster is to split up the image into two or four different sections. Each

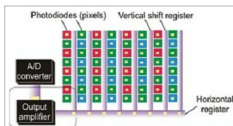


Fig. 4: CCD array with Bayer's filter

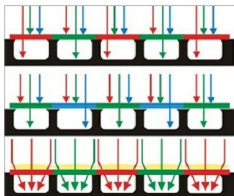


Fig. 5: Colour discrimination by Bayer's filter

section follows the process of parallel and serial shift. Fig. 2 shows a typical CCD sensor with a single-point read-out. Fig. 3 shows a typical packaged CCD chip.

The CCD sensor described in previous paragraphs can only determine number of photons collected by each pixel, and therefore it carries no information about the wavelength or colour of those photons. As a result, the CCD sensor records the image only in monochrome.

In order to record images in full colour, a filter array is bonded to the sensor substrate. One such common colour filter array is the Bayer filter. Bayer's colour filter array (CFA) comprises an arrangement of red, green and blue filters to capture colour information. It is made up of alternating rows of red/green and blue/green filters and is sometimes called an RGBG filter.

Fig. 4 shows a CCD array with a Bayer's filter bonded to its surface. A particular colour filter allows photons of only that colour to pass through to the pixel. This is illustrated in Fig. 5. The number of photons collected by each pixel in this case corresponds to the colour allowed by the filter above it. In Bayer pattern filter, there are twice as many green squares as red or blue squares. This is because the

SINCE 1962

TRMS 9999 Counts DMM
High on Features
Low on Price ...

126B + TRMS

135B + TRMS

Pocket Size

- 4 Digit 9999 Counts
- TRMS Auto / Manual Ranging
- LCD With Backlight
- Upto 999.9V DC
- Upto 750V AC - TRMS
- Upto 9.999A DC / AC - TRMS
- 99.99Ω - 9.999MΩ (126B+TRMS)
- 99.99Ω - 99.99MΩ (135B+TRMS)
- 9.99nF - 9.99mF
- 99.9Hz - 9.99MHz
- 0.1% - 99%
- -20°C - 1000°C (-4°F - 1832°F) (135B+TRMS)
- K Type Thermocouple (Upto 260°C) (135B+TRMS)
- Non-Contact Voltage - NCV (135B+TRMS)
- Square Wave Output 50Hz - 5000Hz (126B+TRMS)

Special Functions
(126B+TRMS & 135B+TRMS)

- Diode test, Audible Continuity, Data Hold, Auto Power Off & Low Battery Indicator

Meco Meters Private Limited
 Plot No. E1-50, MIDC Electronic Zone,
 TTC Industrial Area, Mahape,
 Navi Mumbai - 400710 (INDIA)
 Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
 Fax : 0091-22-2767 3310
 Email : sales@mecomt.com
 Web : www.mecomt.com
 Connect With Us :